From Random Walks to Sequential Testing

The Complete Mathematical Journey in 14 Acts

The story. A test where you can peek at results as often as you like and the false positive guarantee still holds. Getting there requires some beautiful mathematics, built up one step at a time: from coin flips to random walks, from gambling to martingales, from likelihood ratios to Ville's inequality, from Wald's SPRT to modern confidence sequences.

Format. Fourteen interactive acts (Acts 0–13) — deliberately small steps. Each act has an intuitive explanation, mathematical formulation, and key takeaway. Every formula is accompanied by a plain-language translation.

Target audience. People comfortable with algebra and basic probability (e.g. a 12th-grade math background). No calculus is required to follow the main ideas.

Design philosophy. Better to over-explain than under-explain. When in doubt, slow down.

Act 0
The Peeking Problem

Act 1
Random Walk

Act 2
Martingale

Act 3
Likelihood Ratio

Act 4
LR is a Martingale

Act 5
Ville's Inequality

Act 6
Wald's SPRT

Act 7
Mixture / mSPRT

Act 8
Confidence Sequences

Act 9
Problem Eppo Solves

Act 10
Eppo Pipeline

Act 11
Variance Reduction

Act 12
Sequential CI

Act 13
DIY Alternatives

Act 0 — Why Should You Care? The Peeking Problem

The Motivation

Intuitive Explanation

You work at a company running an A/B test. Half your users see the old website, half see a new design. Every day, you check the dashboard: "Is the new design better?" After a week, the dashboard shows $p = 0.03$ — that looks significant! You call it: the new design wins.

But here is the dirty secret of traditional statistics: **the more often you check, the more likely you are to see a false positive.** A test designed to have a 5% false positive rate can easily reach 20–30% if you peek at it repeatedly. One in four or five "winning" experiments might actually be noise.

This is not a theoretical concern. It is the single most common statistical mistake in online experimentation.

Simulation

What is being simulated: 1,000 A/B experiments where there is *no real effect* — any apparent difference is pure noise. Two groups of observations (control and treatment) are generated from *identical* distributions. A standard t-test is computed after every new observation arrives.

A bar chart accumulates: "Fraction of experiments that showed $p < 0.05$ at *some* point during peeking." For 1,000 simulated experiments with regular peeking, this fraction climbs well above 5% — often to 20–30%.

Peeking Simulation

Run 500 A/A experiments (no real effect) and count how often you get a false positive under different peeking schedules.

Run Simulation

Why Does This Happen?

Intuitive Explanation

Traditional statistical tests promise: "If there is no real effect, I will falsely cry wolf at most 5% of the time." But this promise comes with fine print: **you may only look at the result once, at a pre-determined time.**

Every time you peek, you get another chance for randomness to fool you. It is like rolling a die repeatedly — the more rolls, the more likely you are to get a six eventually, even though any single roll has only a $1/6$ chance.

The rest of this presentation builds a test that does not have this problem: a test where you can peek as often as you like and the false positive guarantee still holds.

That promise has a name: **anytime-valid inference**. Getting there requires some beautiful mathematics, which we will build up one step at a time.

How Bad Is It?

Checking schedule	Actual false positive rate
Once at the end (as designed)	$\sim 5\%$
Daily for 1 week	$\sim 13\%$
Daily for 2 weeks	$\sim 19\%$
Daily for 4 weeks	$\sim 25\%$
Continuously (every observation)	Can exceed 30%

Key Takeaway

The peeking problem: Checking a traditional test repeatedly inflates the false positive rate far beyond its nominal level. We need a fundamentally different kind of test — one that remains valid no matter when or how often we look.

Act 1 — The Random Walk

Simulation

A dot starts at position 0. Each step, a coin is flipped. Heads: the dot moves up one step. Tails: down one step. The path is drawn as a graph (position vs. step number).

Random Walk Simulation

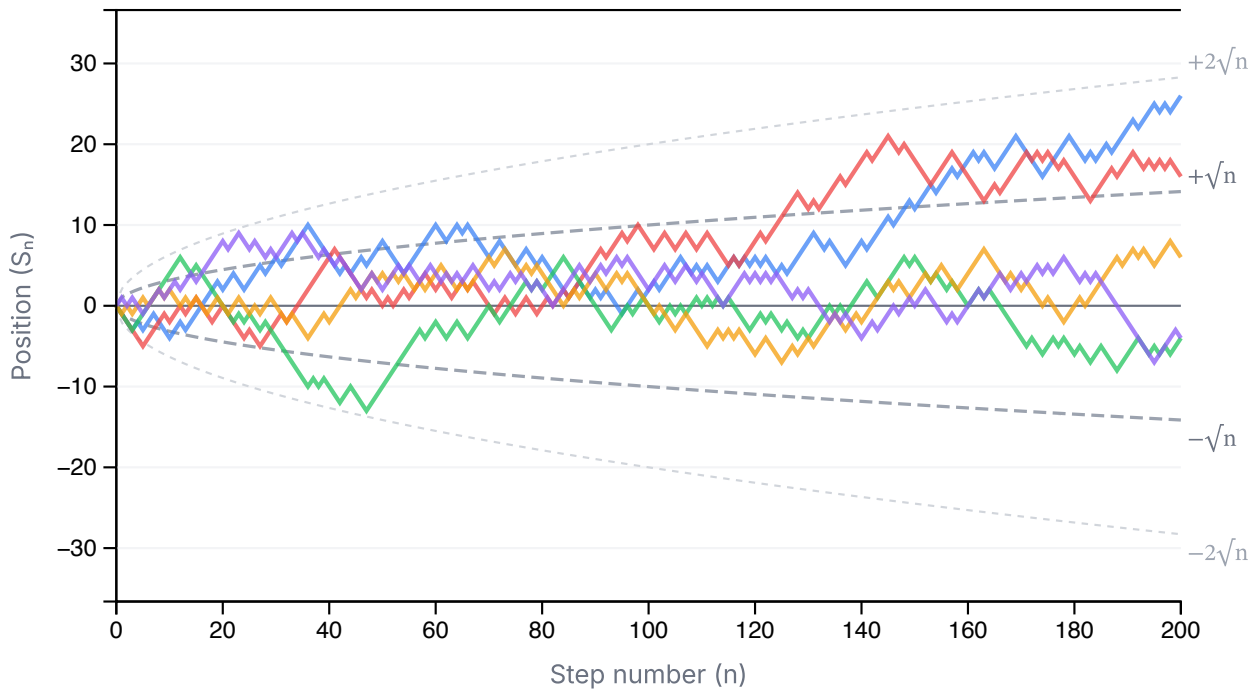
Watch coin-flip random walks fan out over time. Dashed lines show the ±√n and ±2√n envelopes.

New Walks

Steps: 200

P(heads): 0.50

Paths: 5



What you're seeing: Each path is a cumulative sum of ±1 coin flips. With a fair coin (p = 0.5), the expected position is always zero. About 68% of the time the walk stays within $\pm\sqrt{n}$ of zero, and about 95% within $\pm 2\sqrt{n}$.

Intuitive Explanation

Intuitive Explanation

Imagine you are standing in the middle of a long hallway. Flip a coin. Heads — step forward. Tails — step back. Repeat hundreds of times.

Where will you end up? We don't know exactly, but we can say something precise about the *range* of likely outcomes. Most of the time you won't wander far. Occasionally, a long streak of heads or tails carries you far in one direction. The longer you walk, the wider the range of possible positions — but the centre of that range stays at zero.

When we run many people simultaneously, they fan out over time — like ink dropped into water. This spreading has a precise mathematical shape.

Mathematical Formulation

Setting up the notation

Let X_i represent the outcome of the ith coin flip:

$$X_i = \begin{cases} +1 & \text{if heads (step forward)} \\ -1 & \text{if tails (step backward)} \end{cases}$$

The probability of heads is written p. For a fair coin, $p = 0.5$.

Position after n steps

$$S_n = \sum_{i=1}^{n} X_i = X_1 + X_2 + \cdots + X_n$$

Read as: "S sub n equals the sum of all X_i, starting from $i = 1$ up to $i = n$."

Worked example. First 5 flips: H, T, H, H, T, so $X_1 = +1,\ X_2 = -1,\ X_3 = +1,\ X_4 = +1,\ X_5 = -1$.

$$S_1 = 1, \quad S_2 = 0, \quad S_3 = 1, \quad S_4 = 2, \quad S_5 = 1$$

The average outcome (expected value)

When $p = 0.5$, each flip has expected value zero:

$$\mathbb{E}[X_i] = \underbrace{0.5}_{P(\text{heads})} \times (+1) + \underbrace{0.5}_{P(\text{tails})} \times (-1) = 0$$

Because each step averages to zero:

$$\mathbb{E}[S_n] = \mathbb{E}[X_1] + \mathbb{E}[X_2] + \cdots + \mathbb{E}[X_n] = 0$$

On average, across many trials, you end up back where you started.

How spread out the outcomes are — variance & standard deviation

Variance measures the average squared distance from the mean. For a single fair coin flip:

$$\mathrm{Var}(X_i) = \mathbb{E}[(X_i - 0)^2] = 0.5 \times (+1)^2 + 0.5 \times (-1)^2 = 1$$

Because flips are independent, the variance of the sum is the sum of the variances:

$$\mathrm{Var}(S_n) = \mathrm{Var}(X_1) + \mathrm{Var}(X_2) + \cdots + \mathrm{Var}(X_n) = \underbrace{1 + 1 + \cdots + 1}_{n \text{ terms}} = n$$

The **standard deviation** is the square root of the variance — same units as the position:

$$\mathrm{SD}(S_n) = \sqrt{\mathrm{Var}(S_n)} = \sqrt{n}$$

Before we begin flipping the coin, the expected position is zero: $\mathbb{E}[S_n] = 0$. But positions spread out over time. If we flip the coin a large number of times, about 68% of walks would end up within $\pm\sqrt{n}$ of zero (one standard deviation), and about 95% within $\pm 2\sqrt{n}$ (two standard deviations). More generally, the distribution of the sum of walks would be approximately normal. For example, after 100 steps, 68% of walks are within ±10 and 95% within ±20. The spread grows with the *square root* of the number of steps, not the number itself.

What if the coin is not fair?

When $p \neq 0.5$, each step has a non-zero average:

$$\mathbb{E}[X_i] = p \times (+1) + (1 - p) \times (-1) = 2p - 1$$

For $p = 0.6$: $\mathbb{E}[X_i] = 0.2$. After n steps:

$$\mathbb{E}[S_n] = n(2p - 1)$$

This is a systematic *drift* — the random walk now has a trend.

Key Takeaway

Key concepts: stochastic process, expected value, variance, standard deviation, the $\sqrt{n}$ growth of randomness.

Act 2 — The Gambling Game and the Martingale

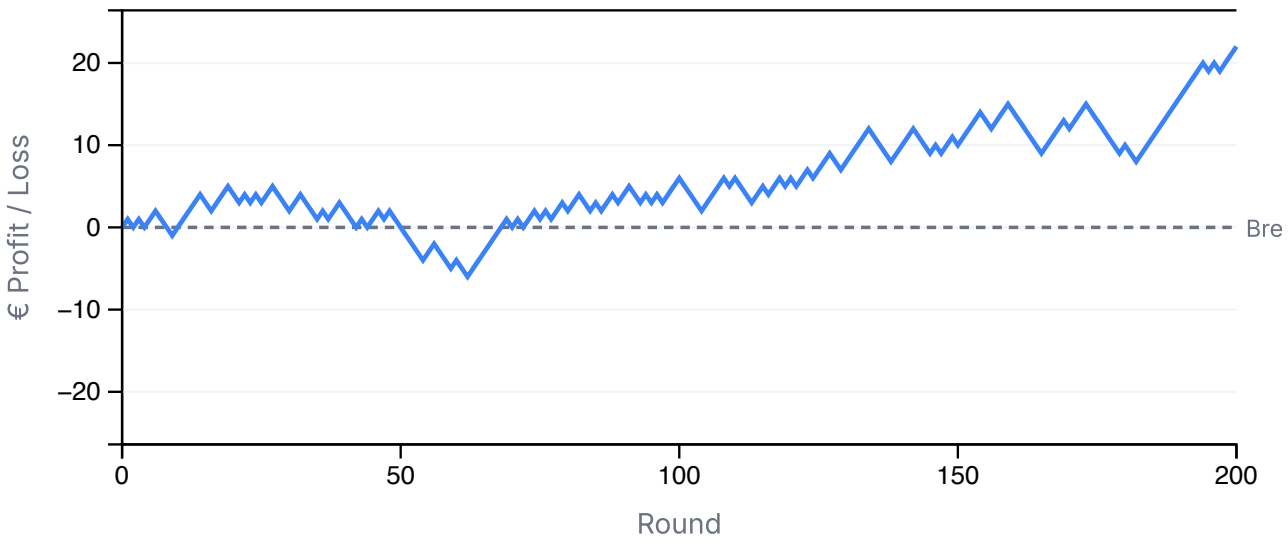
Simulation

The same random walk, relabelled. The vertical axis now shows "€ profit or loss." A bold horizontal line marks zero — break-even.

Doubling strategy mode: Runs 10,000 gamblers using the martingale betting strategy. A histogram shows the distribution of final wealth — many small winners, a few catastrophic losers. A counter shows average profit converging to zero.

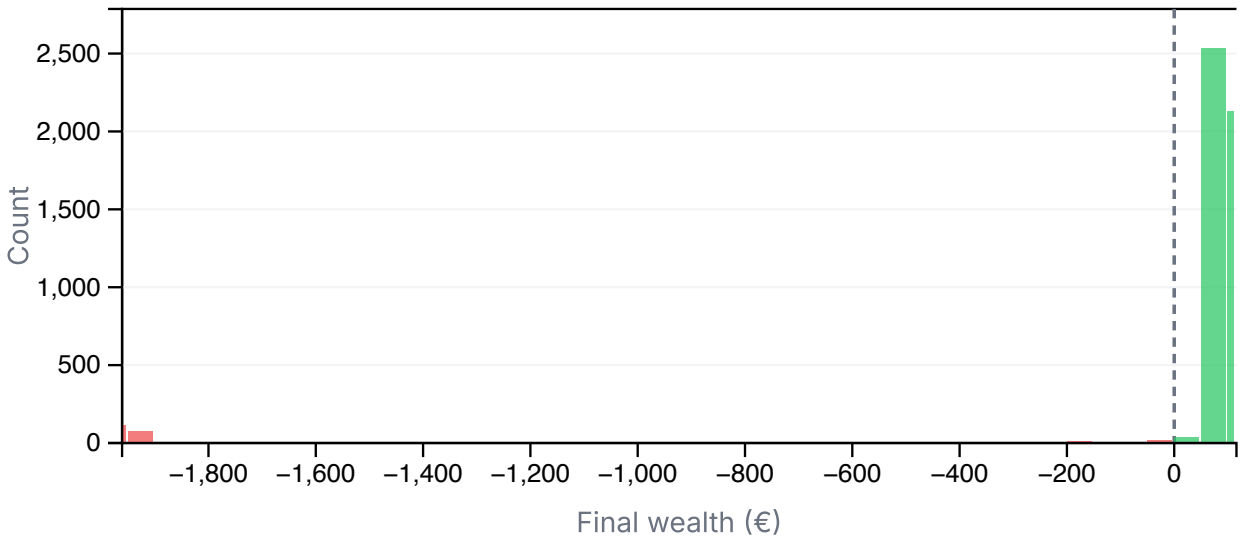
Gambling Game & Doubling Strategy

Fair-coin gambling walk (€1 per flip)



New Walk

Doubling strategy: 5,000 gamblers, 200 rounds each



Re-run Doubling

Gamblers: 5,000

Max rounds: 200

4,734	266	€-3.17
Winners (94.7%)	Losers (5.3%)	Avg profit

What you're seeing: Left: a single gambler's profit/loss over time — a random walk re-labelled as money. Right: the doubling strategy histogram — many small winners, a few catastrophic losers. The average profit converges to €0 (martingale property). No strategy can beat a fair game.

Intuitive Explanation

Intuitive Explanation

Win €1 for heads, lose €1 for tails. Cumulative winnings trace the same random walk from Act 1.

The gambler's temptation: “I'll keep playing until I'm ahead, then quit.” Sounds foolproof — surely a run of luck will eventually put you in the green?

Mathematically, a fair random walk does eventually return to positive territory. But the time it takes can be astronomically long, and while you wait you might lose everything.

Central insight: There is no strategy that improves your expected outcome in a fair game. This is a proven mathematical theorem. Try any stopping rule — the average winnings always converge to zero.

The doubling strategy (Martingale betting), exposed

Lose, double the bet, repeat until you win back your losses plus €1. It works — most of the time.

But a long losing streak forces bets of €1, 2, 4, 8, 16, 32, 64, ... After just 10 consecutive losses, you must bet €1,024 to win back €1. The histogram makes it vivid: a tall bar of people who won €1, and a barely visible bar at the far left of people who lost thousands. Multiply heights by positions and sum: the average is zero.

Mathematical Formulation

Definition: Martingale

A sequence $M_0, M_1, M_2, \ldots$ is a **martingale** if, at every step, the best prediction of the next value is the current value:

$$\mathbb{E}[M_n \mid M_0, M_1, \ldots, M_{n-1}] = M_{n-1}$$

Reading the notation carefully:

- $\mathbb{E}[\cdot]$ = "the expected value of..." (long-run average).
- The vertical bar " $\mid$ " = "given that we know." Everything after the bar is information we have.
- $M_0, M_1, \ldots, M_{n-1}$ = the full history up to step $n-1$.
- Full sentence: "The expected value of M_n, given that we know everything up to step $n-1$, equals M_{n-1}."

Knowing the entire history, your best guess for the next value is just the current value. The past gives no useful information about the future direction.

Verifying: the coin-flip game is a martingale

Let $M_n = S_n$ (cumulative winnings). Then $M_n = M_{n-1} + X_n$.

$$\begin{aligned}
\mathbb{E}[M_n \mid M_0, \ldots, M_{n-1}] &= \mathbb{E}[M_{n-1} + X_n \mid M_0, \ldots, M_{n-1}] \\
&= M_{n-1} + \mathbb{E}[X_n \mid M_0, \ldots, M_{n-1}] \quad (M_{n-1} \text{ is known}) \\
&= M_{n-1} + 0 \quad (\text{next flip is independent; } \mathbb{E}[X_n] = 0) \\
&= M_{n-1} \quad \checkmark
\end{aligned}$$

Doob's Optional Stopping Theorem (1953)

Theorem (Doob's Optional Stopping Theorem)

Let $\{M_n\}$ be a martingale and τ a **stopping time** — a rule for when to quit that uses only information accumulated so far (no looking into the future). Then, under reasonable conditions:

$$\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$$

For a gambler starting with $M_0 = 0$: expected winnings at stopping = 0, always, for any strategy. **You cannot beat a fair game by choosing when to quit.**

What is a stopping time?

τ is a decision rule of the form: “Stop after step n if [condition based only on what happened so far].”

Rule	Valid?
Stop after exactly 100 flips	✓
Stop the first time I am ahead by €10	✓
Stop when I have lost €50	✓
Stop after 1000 flips *or* when ahead by €5, whichever first	✓
Stop right before the next loss	✗ (requires the future!)

The “reasonable conditions” for Doob's theorem

Doob's theorem requires at least one of:

1. $\tau \leq N$ for some fixed N (bounded stopping time), **or**
2. $|M_n| \leq C$ for some fixed C (bounded martingale), **or**
3. $\mathbb{E}[\tau] < \infty$ and certain integrability conditions (finite expected stopping time).

When *none* hold — as with the doubling strategy — the guarantee breaks down.

Why the doubling strategy fails

The doubling strategy guarantees eventual profit *only if* you have infinite money and infinite time. In practice:

- Finite budget ⇒ real probability of ruin before recovery.
- Expected time to recovery is infinite: $\mathbb{E}[\tau] = \infty$.
- Doob's theorem gives $\mathbb{E}[M_\tau] \leq 0$, not $= 0$.

Why this matters for what is coming

The martingale concept reappears in every subsequent act. The likelihood ratio (Act 4) is a martingale. The mixture likelihood ratio (Act 7) is a martingale. The reason sequential tests control false positives — even with peeking — is that they exploit the same structure: **a fair game cannot be beaten by choosing when to stop.**

Key Takeaway

Key concepts: martingale, conditional expectation, stopping time, Doob's Optional Stopping Theorem, the “reasonable conditions.”

Historical Note

Joseph L. Doob formalised martingale theory in his 1953 monograph *Stochastic Processes*. The name “martingale” comes from a French term for a type of betting strategy — the very doubling strategy we just debunked.

Act 3 — Hypothesis Testing: Telling Two Coins Apart

Simulation

The user is presented with a coin and must decide: fair or biased? Two panels run side by side — one with a fair coin, one with a biased coin. For small bias, the sequences look nearly identical for many flips.

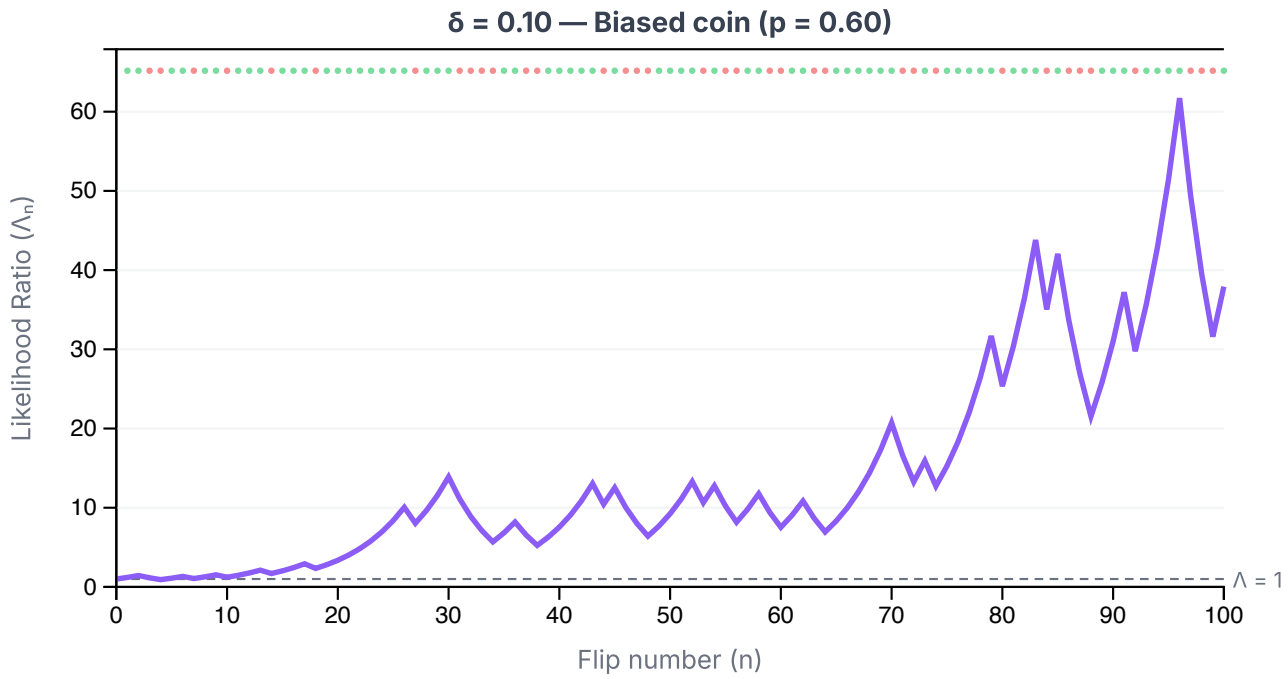
Likelihood Ratio Accumulation

Watch the LR grow as evidence accumulates flip by flip. Green dots = heads, red dots = tails.

New Flips

Bias δ: 0.10 → coin p = 0.60

Flips: 100



37.92	64/100	Favours H_1
Final Λ_n	Heads	Strong evidence

Intuitive Explanation

Intuitive Explanation

You suspect a coin might be unfair. Flip it many times and look at the results. 10 flips with 6 heads? Could be luck. 1,000 flips with 600 heads? Strong evidence.

Statisticians frame this as two competing stories:

- **Null hypothesis H_0:** "The coin is fair. Nothing is happening."
- **Alternative hypothesis H_1:** "The coin is biased. Something real is going on."

Two types of mistakes:

1. **False positive** (Type I): conclude biased when it is fair.
2. **False negative** (Type II): conclude fair when it is biased.

Traditionally: $\alpha = 0.05$ (max false positive rate), $\beta = 0.20$ (max false negative rate). **Power** $= 1 - \beta = 0.80$.

Mathematical Formulation

The likelihood

Suppose n flips produce k heads.

Under H_0 (fair coin, $p = 0.5$):

$$\mathcal{L}_0 = (0.5)^k \times (0.5)^{n-k} = (0.5)^n$$

Under H_1 (biased coin, $p = 0.5 + \delta$):

$$\mathcal{L}_1 = (0.5 + \delta)^k \times (0.5 - \delta)^{n-k}$$

The likelihood ratio

$$\Lambda_n = \frac{\mathcal{L}_1}{\mathcal{L}_0} = \frac{(0.5 + \delta)^k \, (0.5 - \delta)^{n-k}}{(0.5)^n}$$

"How many times more likely is our observed data under the biased-coin story compared to the fair-coin story?"

Worked example. $\delta = 0.1$ (biased coin has $p = 0.6$); $n = 10$ flips, $k = 7$ heads.

$$\mathcal{L}_1 = (0.6)^7 \times (0.4)^3 \approx 0.001792$$
$$\mathcal{L}_0 = (0.5)^{10} \approx 0.000977$$
$$\Lambda_{10} = \frac{0.001792}{0.000977} \approx 1.83$$

The data is ≈ 1.83 times more likely if the coin is biased. Evidence, but not overwhelming.

Interpreting Λ_n

Λ_n	Interpretation
$= 1$	Data equally consistent with both hypotheses
$= 10$	10× more likely under H_1 — moderate evidence
$= 100$	100× more likely under H_1 — strong evidence
$= 0.1$	10× more likely under H_0 — evidence for fairness

Incremental updating

Λ_n can be computed one flip at a time:

$$\Lambda_n = \Lambda_{n-1} \times \frac{f_1(x_n)}{f_0(x_n)}$$

where $f_j(x_n)$ is the probability of the nth flip's outcome under H_j.

- Heads: multiply by $\frac{0.5+\delta}{0.5}$.
- Tails: multiply by $\frac{0.5-\delta}{0.5}$.

This incremental structure is what makes *sequential* testing possible.

Starting value

Before any data: $\Lambda_0 = 1$. Neither hypothesis is favoured. This will be important for Ville's inequality (which requires the starting value to equal 1).

Key Takeaway

Key concepts: null hypothesis, alternative hypothesis, likelihood, likelihood ratio, false positive, false negative, power, incremental updating.

Act 4 — The Likelihood Ratio Is a Martingale

Simulation

The likelihood ratio Λ_n is plotted as a path over time — like the random walk from Act 1.

Left panel: 100 paths of Λ_n when the coin is truly *fair* (H_0 true). Paths wander with no systematic trend.

Right panel: 100 paths when the coin is truly *biased*(H_1 true). Paths drift upward.

LR Paths: H_0 vs H_1

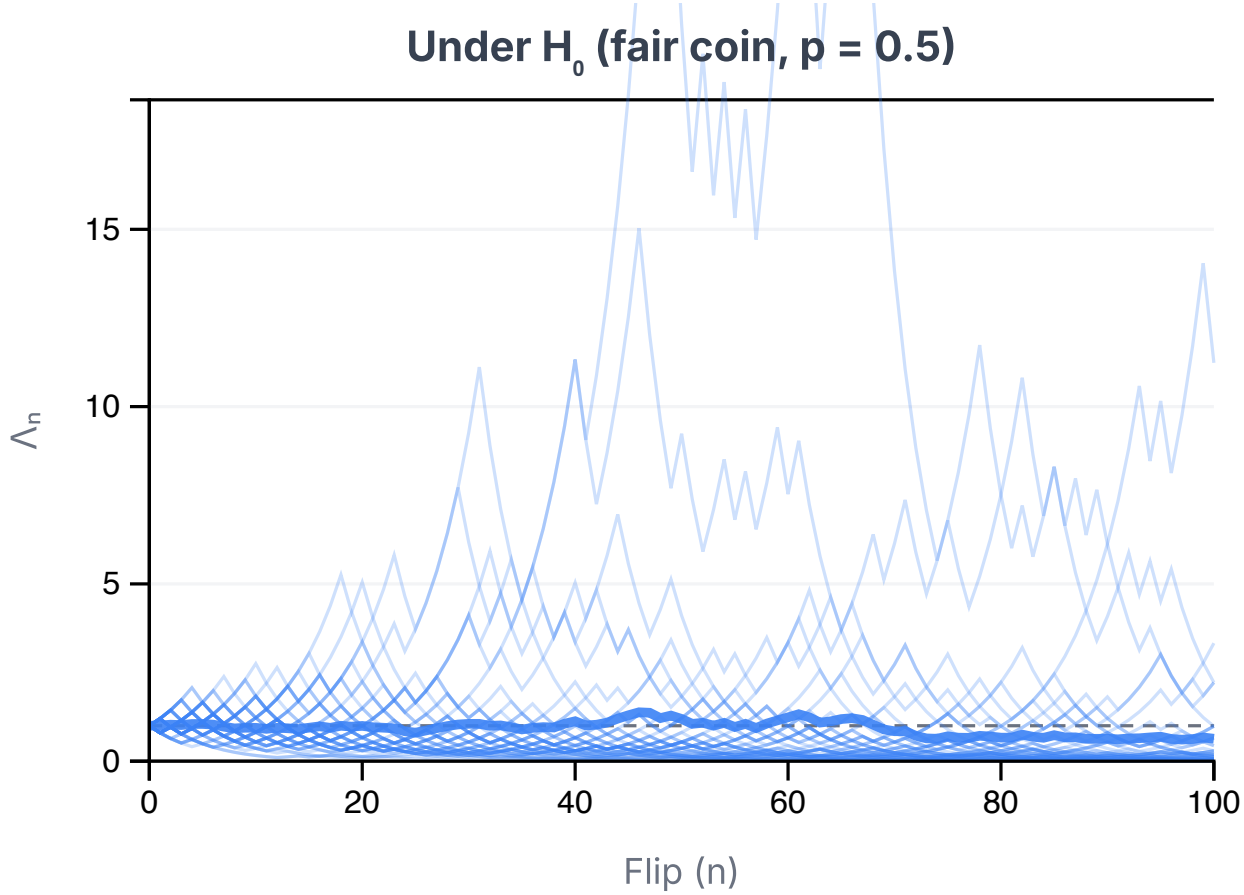
50 likelihood ratio paths. Under H_0 they wander (martingale); under H_1 they drift up. Bold line = average path.

Regenerate

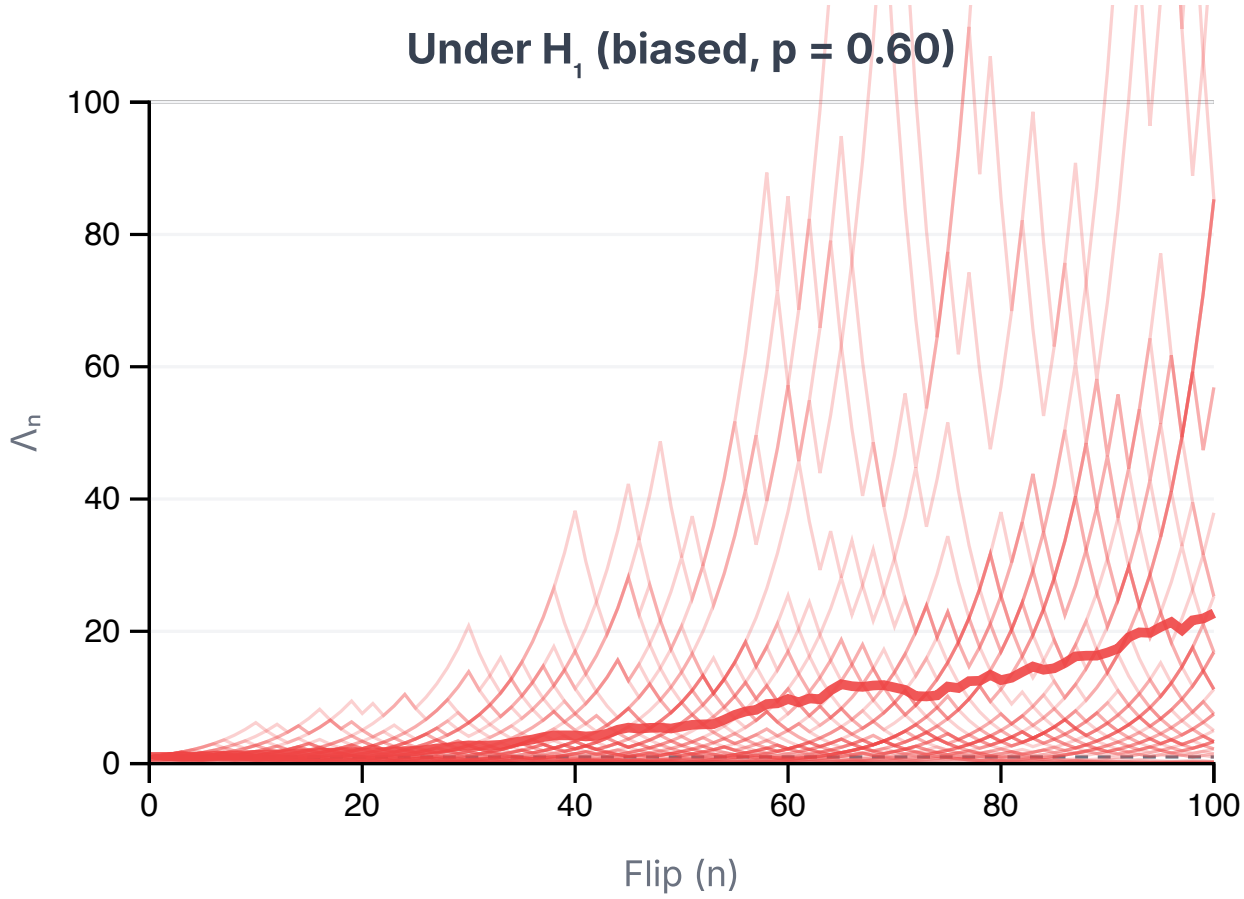
δ: 0.10

Flips: 100

Paths: 50



Mean final Λ: 0.63 (should ≈ 1)



Mean final Λ: 22.73 (drifts upward)

Intuitive Explanation

Intuitive Explanation

Under H_0, the likelihood ratio wanders up and down but never develops a consistent trend. On average, it stays at 1.

This *is* a martingale — exactly like the gambler's cumulative winnings. Everything from Act 2 applies: Doob's theorem says you cannot systematically make Λ_n large by choosing clever peeking times.

Mathematical Formulation

Proof that Λ_n is a martingale under H_0

We must show: $\mathbb{E}[\Lambda_n \mid \Lambda_0, \ldots, \Lambda_{n-1}] = \Lambda_{n-1}$.

From Act 3, $\Lambda_n = \Lambda_{n-1} \times \frac{f_1(x_n)}{f_0(x_n)}$. Therefore:

$$\begin{aligned}\mathbb{E}[\Lambda_n \mid \text{past}] &= \mathbb{E}\left[\Lambda_{n-1} \cdot \frac{f_1(x_n)}{f_0(x_n)} \;\middle|\; \text{past}\right] \\ &= \Lambda_{n-1} \cdot \mathbb{E}\left[\frac{f_1(x_n)}{f_0(x_n)}\right] \quad (\Lambda_{n-1} \text{ is known})\end{aligned}$$

Now the key step. Under H_0, x_n is drawn from f_0. So:

$$\begin{aligned}\mathbb{E}\left[\frac{f_1(x_n)}{f_0(x_n)}\right] &= \sum_{\text{all outcomes } x} f_0(x) \cdot \frac{f_1(x)}{f_0(x)} \\ &= \sum_x \cancel{f_0(x)} \cdot \frac{f_1(x)}{\cancel{f_0(x)}} \quad \textbf{(the cancellation!)} \\ &= \sum_x f_1(x) \\ &= 1 \quad (f_1 \text{ is a probability distribution})\end{aligned}$$

Therefore:

$$\mathbb{E}[\Lambda_n \mid \text{past}] = \Lambda_{n-1} \times 1 = \Lambda_{n-1} \quad \checkmark$$

The cancellation with actual numbers

For $\delta = 0.1$ (biased coin $p = 0.6$):

Outcome	$f_0(x)$	$f_1(x)/f_0(x)$	$f_0(x) \times [f_1(x)/f_0(x)]$
Heads	0.5	$0.6/0.5 = 1.2$	$0.5 \times 1.2 = 0.6$
Tails	0.5	$0.4/0.5 = 0.8$	$0.5 \times 0.8 = 0.4$
Total:			**1.0 ✓**

When we compute the expected ratio under H_0, the H_0 probabilities cancel out, and we are just left summing H_1 probabilities — which must add to 1. This is the "aha" moment of the proof.

Two additional properties

1. Λ_n **is non-negative:** It is a product of non-negative ratios (probabilities divided by probabilities), so $\Lambda_n \geq 0$.
2. $\Lambda_0 = 1$**:** Before any data, neither hypothesis is favoured.

So Λ_n is a **non-negative martingale starting at 1**. Remember these properties — they are exactly what Ville's inequality needs.

Under H_1, Λ_n drifts upward

When the coin really is biased:

$$\mathbb{E}\left[\frac{f_1(x_n)}{f_0(x_n)}\right]_{H_1} = \sum_x f_1(x) \cdot \frac{f_1(x)}{f_0(x)} > 1$$

So Λ_n tends to grow — this upward drift is the signal the test detects.

Key Takeaway

Key concepts: the likelihood ratio is a martingale under H_0; the cancellation that makes it work; non-negative martingale; drift under H_1.

Act 5 — Markov, Ville, and the Anytime-Valid Guarantee

This is the pivotal act of the entire presentation.

Simulation

Panel 1 — Markov (single time): 10,000 fair-coin paths of Λ_n. A threshold at $\Lambda = 1/\alpha = 20$. Count: how many paths are above the threshold at step 200? Fraction $\leq 5\%$.

Panel 2 — Ville (any time): Same paths. Count: how many *ever* crossed the threshold at *any* step? Also $\leq 5\%$ — but requires a stronger theorem.

Panel 3 — Comparison: Standard z-score checked at every step. Count how many ever show significance. Much higher than 5% — the peeking problem.

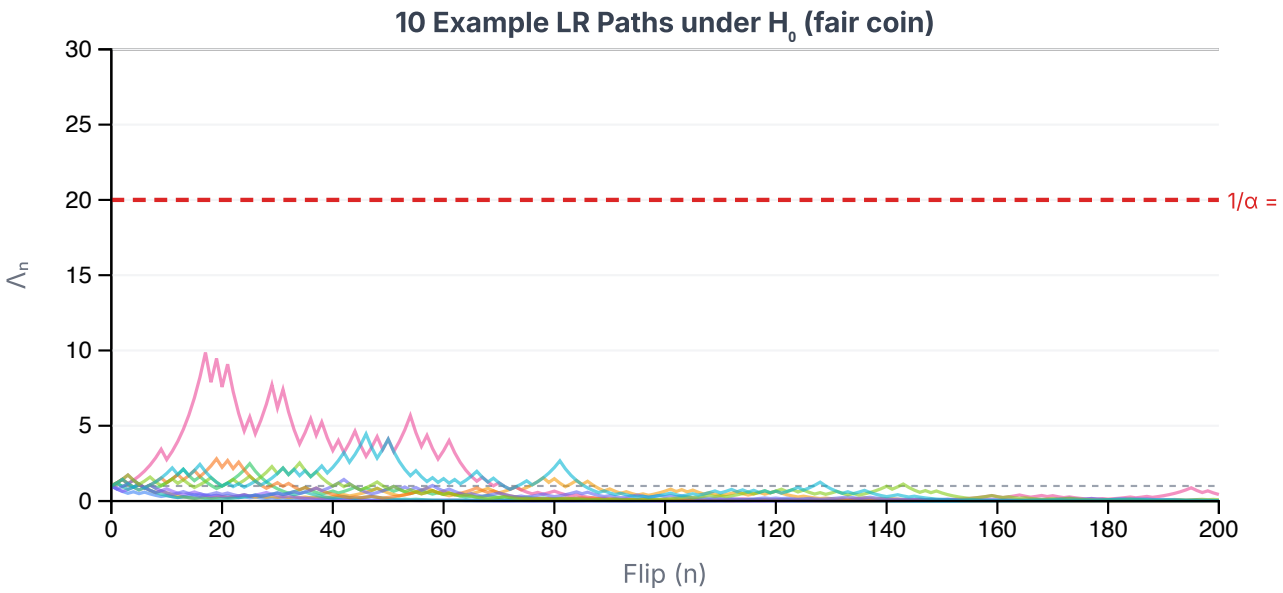
Markov vs Ville vs Peeking z-test

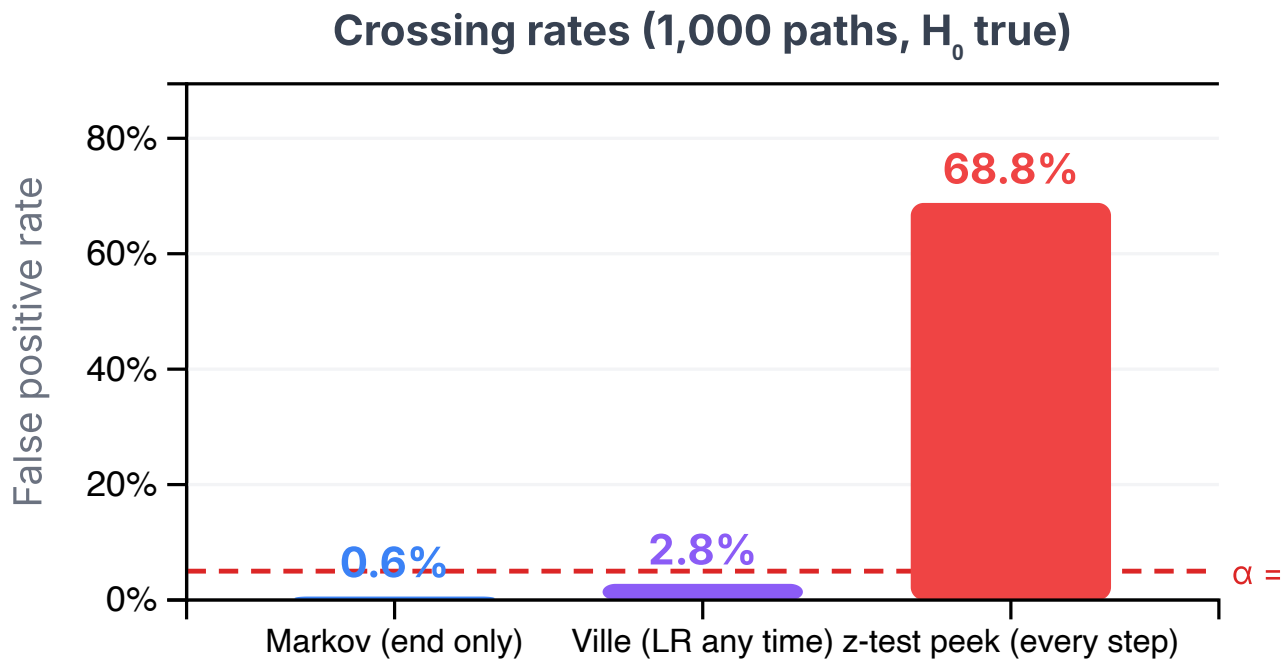
All under H_0 (fair coin). Compare: crossing threshold at one time (Markov), ever crossing (Ville), and repeated z-test significance.

Re-run

Paths: 1,000

Flips: 200





Key insight: Both Markov (0.6%) and Ville (2.8%) stay below α = 5%, while the standard z-test with peeking (68.8%) far exceeds it. The LR threshold is anytime-valid; the p-value threshold is not.

Intuitive Explanation

Intuitive Explanation

We are about to prove the most important result in this entire presentation: the **anytime-valid guarantee**.

Two steps:

1. **Markov's inequality** — a simple bound at one point in time.
2. **Ville's inequality** — extends the bound to all points in time simultaneously.

Ville's inequality is *the reason* sequential testing works.

Mathematical Formulation

Step 1: Markov's Inequality

Theorem (Markov's Inequality)

For any non-negative random variable X with finite $\mathbb{E}[X]$:

$$\mathbb{P}(X \geq c) \leq \frac{\mathbb{E}[X]}{c}$$

$\mathbb{P}(\cdot)$ denotes probability — a number between 0 and 1 measuring how likely an event is. So "the probability that X is at least c cannot exceed the average of X divided by c."

Why is this true? An intuitive proof.

Suppose $X \geq 0$ always. Let $q = \mathbb{P}(X \geq c)$ be the fraction of outcomes at or above c. Those outcomes contribute *at least* $q \times c$ to the average (each is $\geq c$). The rest contribute ≥ 0 (since $X \geq 0$):

$$\mathbb{E}[X] \geq q \cdot c + (1-q) \cdot 0 = q \cdot c \quad \Longrightarrow \quad q \leq \frac{\mathbb{E}[X]}{c} \quad \checkmark$$

Concrete example. Average income in a town is €50,000.

$$\mathbb{P}(\text{income} \geq €500{,}000) \leq \frac{50{,}000}{500{,}000} = 0.10 = 10\%$$

At most 10% of people can earn €500,000+.

Applying Markov to the likelihood ratio.

Under H_0, Λ_n is a martingale with $\mathbb{E}[\Lambda_n] = \Lambda_0 = 1$ and $\Lambda_n \geq 0$. Markov gives:

$$\mathbb{P}\left(\Lambda_n \geq \frac{1}{\alpha}\right) \leq \frac{\mathbb{E}[\Lambda_n]}{1/\alpha} = 1 \cdot \alpha = \alpha$$

At any single, pre-specified step n, the false positive probability is at most α. Reassuring — but only for *one* moment.

Step 2: Ville's Inequality (1939)

Theorem (Ville's Inequality)

For any **non-negative martingale** $\{M_n\}_{n\geq 0}$ with $M_0 = m_0$:

$$\mathbb{P}\left(\sup_{n\geq 0} M_n \geq c\right) \leq \frac{m_0}{c}$$

where $\sup_{n\geq 0} M_n$ is the highest value M_n *ever* reaches.

Why this is remarkable:

Markov	At any single *fixed* moment, $\mathbb{P}(M_n \geq c)$ is small.
Ville	$\mathbb{P}(M_n$ **ever** $\geq c$, at **any** moment$)$ is *also* small.

The intuition behind Ville.

For a non-negative martingale, reaching a high value “uses up” the available expected value. Once high, it must stay high on average (martingale property), yet it cannot go below zero (non-negativity). These two constraints together limit how often it can ever be high.

Applying Ville to the likelihood ratio

Set $M_n = \Lambda_n$, $m_0 = \Lambda_0 = 1$, $c = 1/\alpha$:

$$\mathbb{P}\left(\Lambda_n \text{ ever reaches } \frac{1}{\alpha} \text{ or higher}\right) \leq \alpha$$

The probability that the likelihood ratio ever crosses our rejection threshold — at any point, across all flips, even if you run forever — is at most α.

This is the anytime-valid guarantee. You can peek after every observation. The false positive rate, across all checks combined, stays $\leq \alpha$.

Connecting back to Act 0

Standard test statistics (like p-values from a t-test) are *not* martingales. Checking repeatedly gives many independent chances to be fooled.

The likelihood ratio *is* a martingale under H_0. Ville's inequality converts this into a practical guarantee: **peeking is safe.**

Historical Note

Jean Ville proved this inequality in his 1939 doctoral thesis, *Étude critique de la notion de collectif*. It predated the formal development of martingale theory by Doob (1953) and was largely overlooked for decades. Robbins (1970) was among the first to recognise its practical importance for sequential testing — a connection we explore in Act 7.

Key Takeaway

Key concepts: Markov's inequality (with proof), Ville's inequality, the anytime-valid guarantee, martingale + non-negative = peek-safe.

Act 6 — SPRT: Wald's Sequential Probability Ratio Test (1945)

Simulation

Two coins: one fair (H_0), one biased (H_1). The user does not know which they have.

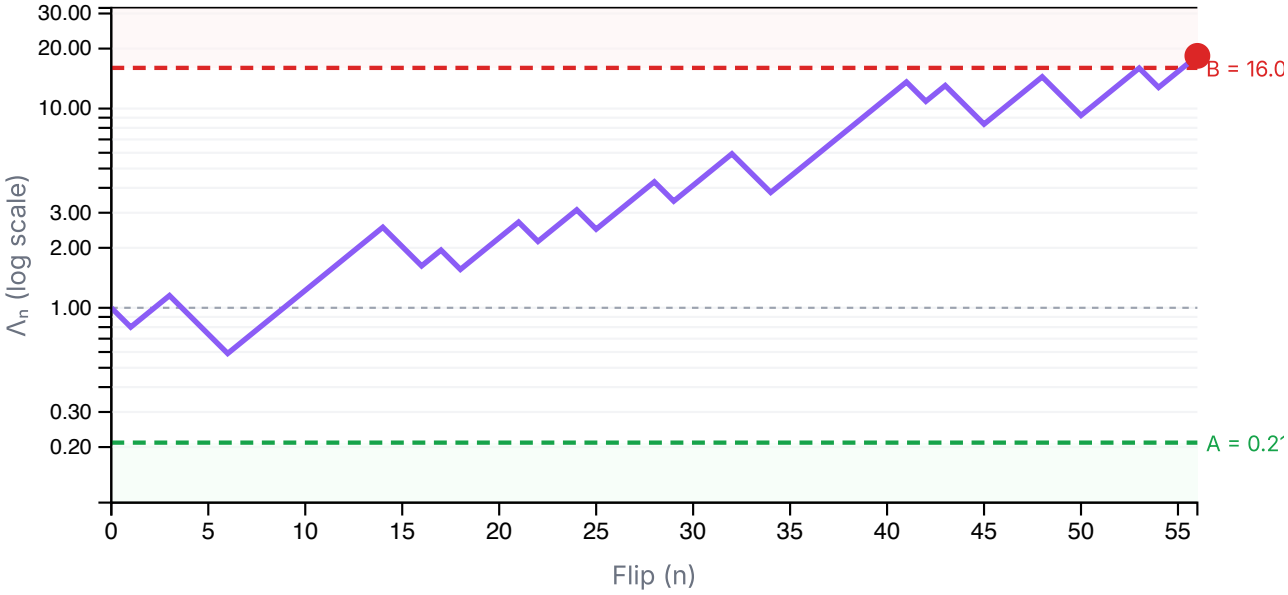
Λ_n is plotted as a path. Two thresholds: **Upper** at $\Lambda_n = B$: cross this ⇒ “coin is biased.” **Lower** at $\Lambda_n = A$: fall below this ⇒ “coin is fair.”

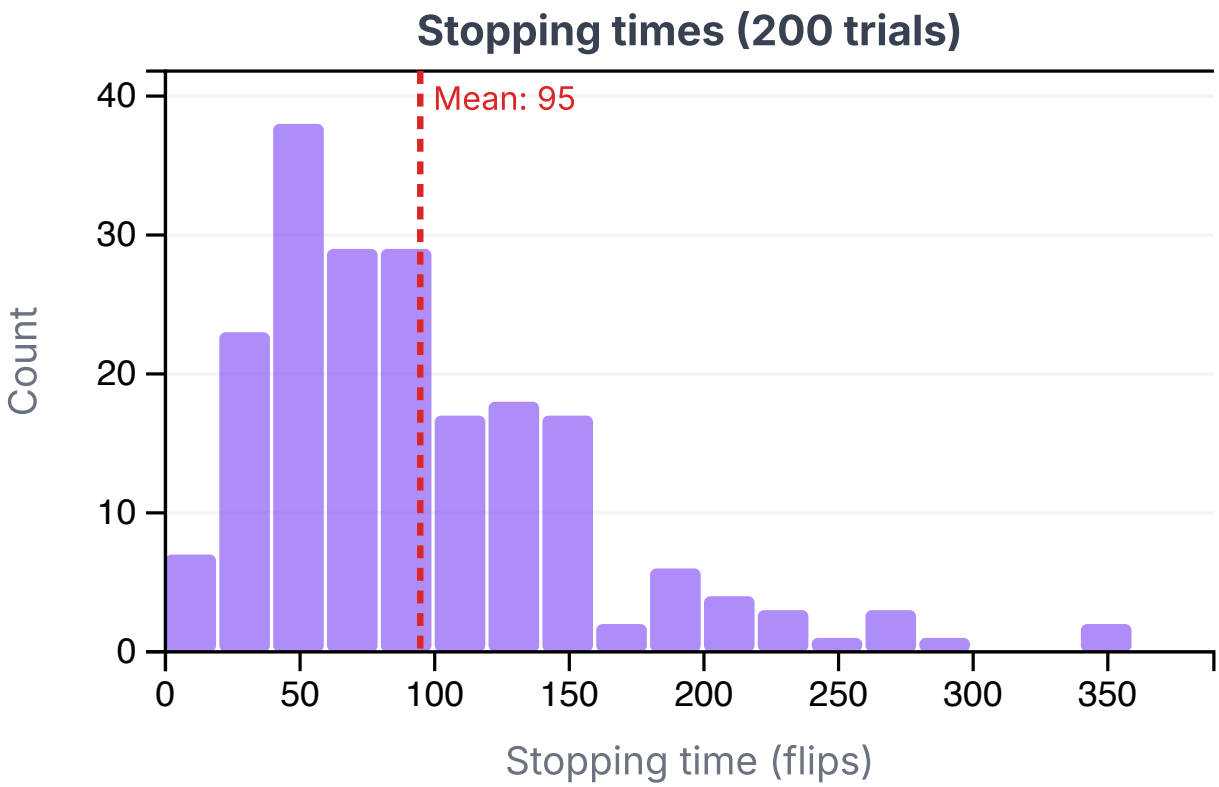
SPRT Decision Boundaries

Re-run

Watch Λ_n traverse between accept (green) and reject (red) boundaries.

δ: 0.10 α: 0.05 β: 0.20 True coin: Biased (p = 0.60





156	44	56
Reject H_0 (78.0%)	Accept H_0 (22.0%)	Example stop: Reject H_0 (biased)

Intuitive Explanation

Intuitive Explanation

Abraham Wald (1943, published 1945) asked: given that we can peek at the data anytime, what is the most *efficient* way to decide between two hypotheses?

His answer: compute Λ_n after each observation. If it gets high enough, conclude H_1. Low enough, conclude H_0. Otherwise, keep collecting.

Wald proved this is **optimal**: no other sequential test with the same error guarantees needs fewer observations on average. In fact, the SPRT typically requires **47–63% fewer observations** than the best fixed-sample test with the same error rates.

Mathematical Formulation

Decision rule

Choose error tolerances α (false positive rate) and β (false negative rate). Compute two thresholds:

$$B = \frac{1-\beta}{\alpha}, \quad A = \frac{\beta}{1-\alpha}$$

Example ($\alpha = 0.05$, $\beta = 0.20$):

$$B = \frac{0.80}{0.05} = 16, \quad A = \frac{0.20}{0.95} \approx 0.211$$

Condition	Decision	Meaning
$\Lambda_n \geq B = 16$	Reject H_0	"Coin is biased"
$\Lambda_n \leq A \approx 0.21$	Accept H_0	"Coin is fair"
$A < \Lambda_n < B$	Keep flipping	"Not enough evidence yet"

Savings over fixed-sample tests

α	β	Saving in expected observations
0.05	0.05	$\approx 53\%$
0.05	0.01	$\approx 47\%$
0.01	0.01	$\approx 63\%$

The SPRT needs, on average, only about **half as many observations** as a traditional fixed-sample test. This was a remarkable result in 1945 and remains one of the main motivations for sequential testing.

The critical weakness

SPRT requires specifying δ in advance — the exact alternative hypothesis H_1. If you set $\delta = 0.1$ but the true bias is 0.03, the likelihood ratio is computed against the wrong alternative. The test may take much longer, or still determine the correct answer but far less efficiently.

In A/B testing — where the effect size is typically unknown — this is a serious limitation. The next two acts address this.

Historical Note

Wald conjectured in his 1945 paper that the SPRT is *exactly* optimal. Three years later, Wald & Wolfowitz (1948) proved the general result: among **all** sequential tests with the same or smaller error probabilities (α, β), the SPRT minimises the expected sample size under both H_0 and H_1 simultaneously.

Key Takeaway

Key concepts: SPRT decision rule, Wald's boundary approximations, 47–63% saving over fixed-sample tests, optimality (Wald & Wolfowitz 1948), the critical dependence on a pre-specified effect size.

Act 7 — The Mixture Approach: From Robbins (1970) to the mSPRT

Two parts: Robbins' key idea, then Johari et al.'s application to A/B testing.

Part A: Robbins' Mixture Approach (1970)

Intuitive Explanation

Wald's SPRT is like betting everything on one horse: you pick a specific effect size δ and compute evidence based on that choice. If the horse wins (the true effect is near δ), you are fast. Otherwise, slow.

Herbert Robbins proposed a radical alternative in 1970: **bet on all horses simultaneously.** Instead of choosing one δ, spread your "evidence budget" across many possible effect sizes using a weighting scheme (a mixing distribution H).

The key insight: this average likelihood ratio is *still* a non-negative martingale under H_0. Ville's inequality still applies. The anytime-valid guarantee is preserved.

The fundamental result

Robbins observed that if Λ_n^δ is the likelihood ratio for a specific effect size δ, then the **mixture likelihood ratio**

$$\Lambda_n^H = \int \Lambda_n^\delta \, dH(\delta)$$

is also a non-negative martingale starting at 1 under H_0, because:

1. Each Λ_n^δ is a non-negative martingale under H_0 (Act 4).
2. A weighted average (integral) of martingales is a martingale (linearity of conditional expectation).

Therefore Ville's inequality gives:

$$\mathbb{P}\left(\Lambda_n^H \text{ ever} \geq \frac{1}{\alpha}\right) \leq \alpha$$

Symbol by symbol:

- Λ_n^δ = the likelihood ratio assuming effect size δ.
- $dH(\delta)$ = how much weight to give effect size δ.
- $\int$ = continuous version of summation — averages over all possible δ.

Think: "average likelihood ratio across an infinitely large expert panel, weighted by credibility."

SPRT vs mSPRT Comparison

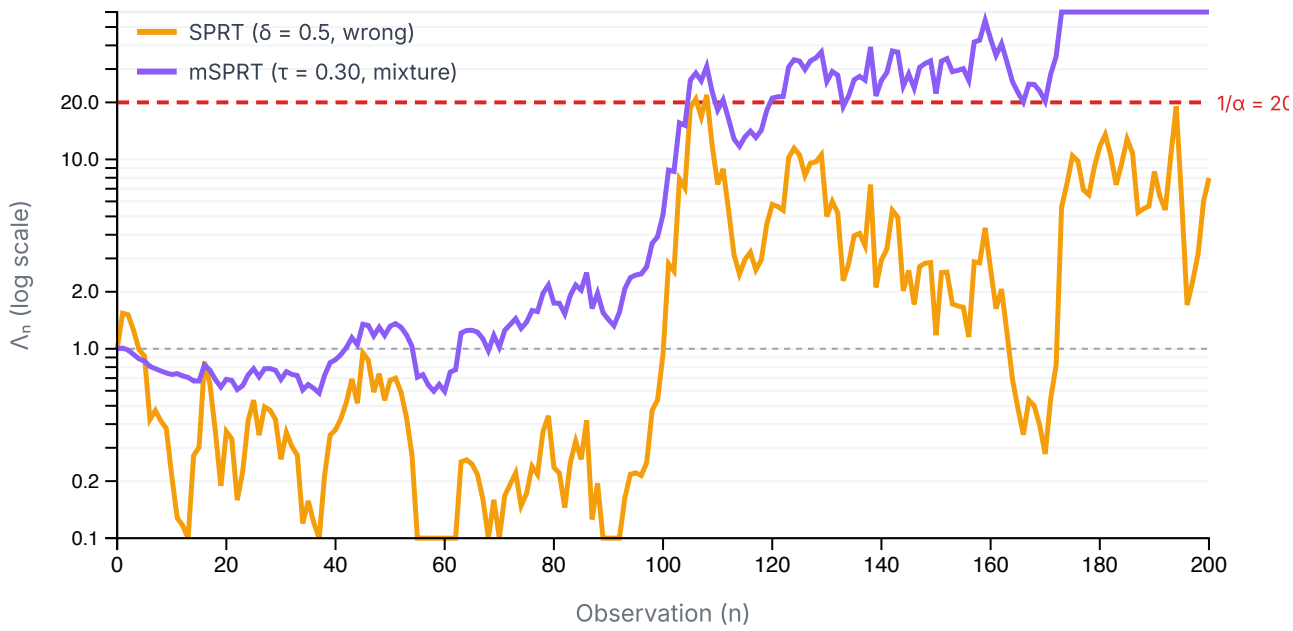
SPRT assumes δ = 0.5 (wrong guess). mSPRT uses a mixture over effect sizes. The true effect is 0.20.

Re-run

True effect: 0.20

Mixture τ: 0.30

Observations: 200



SPRT (wrong δ)
Reject at n=106

mSPRT (mixture)
Reject at n=105

Key insight: When the true effect size differs from the SPRT's assumed δ, the SPRT can be slow or erratic. The mSPRT hedges across many effect sizes via the mixture, making it robust. Both are anytime-valid — the difference is power and speed.

Part B: The mSPRT for A/B Testing (Johari et al., 2017)

Intuitive Explanation

Johari, Pekelis, and Walsh (2017) applied Robbins' idea to the specific setting of A/B tests. They chose the Normal distribution as the mixing distribution H and derived a closed-form formula for the mixture likelihood ratio. They called it the **mixture Sequential Probability Ratio Test (mSPRT)**.

Their key contribution: making the theory practical. They gave explicit formulas, calibration guidance for τ, and proved that being within an order of magnitude of the true effect-size variance is sufficient for near-optimal power.

Closed form for Normal observations

Assumptions:

- Observations $x_i \sim \mathcal{N}(\mu, \sigma^2)$ (Normal with mean μ, **known** variance σ^2).
- Mixing distribution $H = \mathcal{N}(0, \tau^2\sigma^2)$ (Normal centred at 0, variance proportional to σ^2).

The mixture likelihood ratio (Johari et al., Proposition 1):

$$\Lambda_n^H = \frac{1}{\sqrt{1+n\tau^2}} \exp\left(\frac{n\tau^2\,\bar{x}_n^2}{2\,(\sigma^2/n)\,(1+n\tau^2)}\right)$$

Unpacking every piece:

Symbol	Meaning
σ^2	Variance of individual observations (how noisy the data is)
n	Number of observations so far
τ^2	Variance of the mixing distribution; controls how wide a range of effect sizes we consider plausible
$\bar{x}_n$	Sample mean after n observations
$1/\sqrt{1+n\tau^2}$	Shrinkage factor; slowly decreases as n grows

Symbol	Meaning
$\exp(\cdot)$	$e \approx 2.718$ raised to the argument; grows rapidly when $\bar{x}_n$ is far from zero

"The mSPRT score is large when the observed average is far from zero (strong signal), there are many observations, and the noise is low relative to the prior width."

Decision rule

$$\text{Stop and reject } H_0 \text{ when:} \quad \Lambda_n^H \geq \frac{1}{\alpha}$$

For $\alpha = 0.05$: stop when $\Lambda_n^H \geq 20$.

No lower boundary. Unlike the SPRT, the mSPRT cannot definitively confirm H_0. In practice, experiments run for a fixed maximum duration; if the threshold is never crossed, conclude: "No significant effect detected."

Key assumption: known variance

The mSPRT formula assumes σ^2 is known. For continuous data this is typically estimated; for binary data (e.g. click/no-click), the exact mSPRT results hold only approximately via the Central Limit Theorem. For large samples ($n \geq 100$, nearly always true in A/B testing), the approximation is excellent.

Key Takeaway

Key concepts: Robbins' mixture idea, the mSPRT as a weighted average of likelihood ratios, the closed-form Normal formula, the role of τ, no lower boundary, the known-variance assumption.

What's next: The mSPRT was a landmark — but it is not what Eppo uses. The next act introduces **confidence sequences** from Howard et al. (2021), a more general framework that Eppo adopted instead.

Act 8 — Confidence Sequences: The Modern Framework

(Howard et al., 2021)

Simulation

Data arrives one observation at a time. Two bands track the estimated mean: a **fixed 95% CI** (recomputed at each step) — shrinks as $1/\sqrt{n}$, but invalidated by peeking — and a **confidence sequence** — wider at first, but valid at *all* times simultaneously.

Confidence Sequence vs Fixed CI

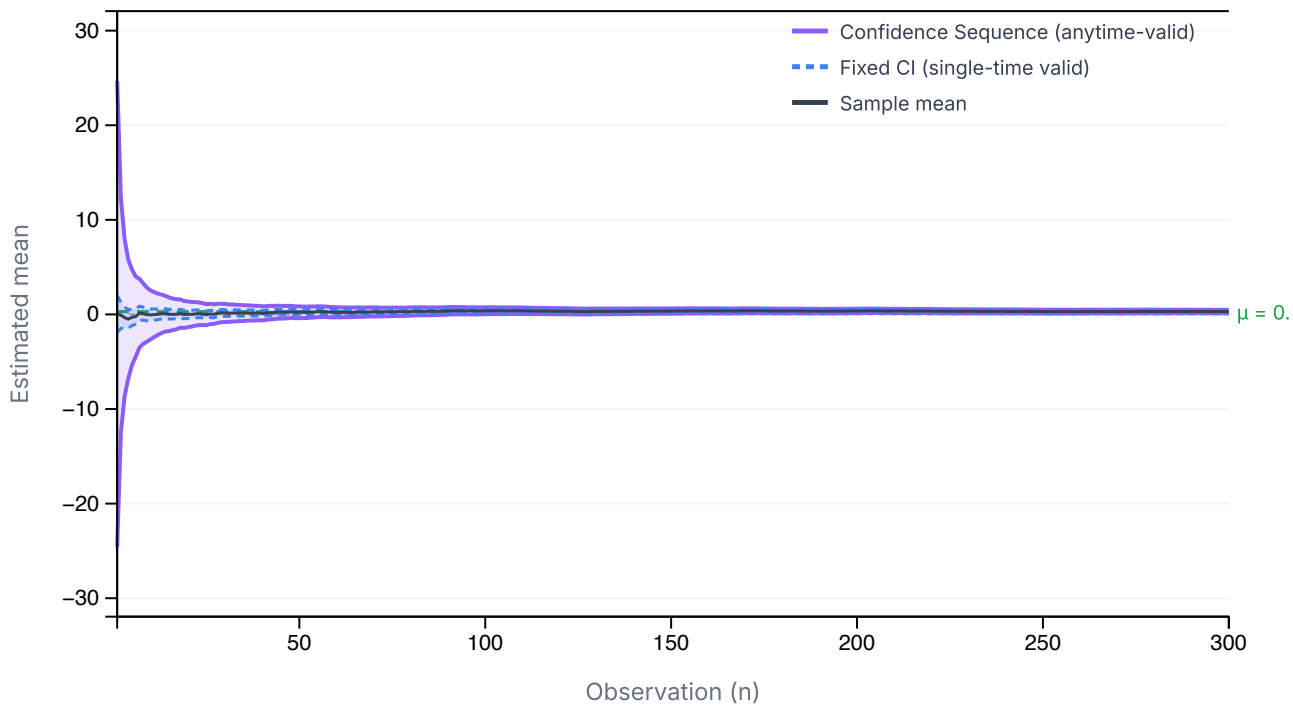
The CS (purple) is wider than the fixed CI (blue) — the "price of peeking" — but valid at all times simultaneously.

Re-run

True effect μ: 0.30

Tuning v: 100

Observations: 300



0.226	0.362	1.60×

Fixed CI width at n=300	CS width at n=300	CS / CI ratio (price of peeking)

Intuitive Explanation

Intuitive Explanation

A confidence sequence is the natural answer to the peeking problem for estimation (not just testing).

A standard 95% confidence interval says: "We are 95% confident the true value is in this range *right now*." But if you check again tomorrow with new data, the guarantee resets. Check 100 times and you are almost guaranteed to be wrong at least once.

A confidence sequence says: "We are 95% confident the true value has been in this range *at every single time we have ever checked, and at every time we will ever check in the future*."

This is a strictly stronger guarantee. The price: the band is wider than a fixed CI at any single time. But in exchange, you are free to monitor continuously, stop whenever you want, and report at any time.

Mathematical Formulation

Definition

Definition (Confidence Sequence)

A sequence of intervals $(C_t)_{t\geq 1}$ is a $(1-\alpha)$-**confidence sequence** (CS) for a parameter μ if:

$$\mathbb{P}\big(\mu \in C_t \text{ for all } t \geq 1\big) \geq 1-\alpha$$

The **entire sequence** of intervals simultaneously covers the true parameter with probability at least $1-\alpha$. Not just at one time — at every time.

The Normal Mixture Boundary (the key formula)

The centrepiece of Howard et al. is the **Normal mixture confidence sequence**. For sub-Gaussian observations with intrinsic time (cumulative variance) $v_t = \sum_{i=1}^{t} \sigma_i^2$:

$$u(v) = \sqrt{(v+\nu) \cdot \log \frac{v+\nu}{\nu\alpha^2}}$$

The $(1-\alpha)$-confidence sequence is:

$$C_t = \left(\bar{x}_t - \frac{u(v_t)}{t},\ \bar{x}_t + \frac{u(v_t)}{t} \right)$$

Unpacking the boundary:

Symbol	Meaning
v	**Intrinsic time**: the cumulative variance of observations so far. For i.i.d. data: $v_t = t\sigma^2$.
$\nu > 0$	**Tuning parameter**: controls which sample sizes the boundary is tightest for. The boundary is approximately optimised for $v \approx \nu$.
α	Significance level (e.g. 0.05)
$\log(\cdot)$	Natural logarithm

The boundary $u(v)$ tells you how far from zero the running sum S_t can drift before you can reject H_0. The parameter ν lets you tune where the boundary is tightest. If you expect to collect N observations total with variance σ^2, setting $\nu \approx N\sigma^2$ makes the boundary tightest around $n = N$.

Connection to Robbins and the mSPRT

Howard et al. explicitly credit Robbins (1970) as their starting point:

- **Robbins (1970):** introduced the mixture approach and confidence sequences, but only for known-variance Normal data.
- **Johari et al. (2017):** made it practical for A/B testing via the mSPRT, but still assumed known variance and parametric (Normal) data.
- **Howard et al. (2021): extended to nonparametric settings** via the sub-ψ framework. The boundary works for *any* sub-Gaussian data (not just Normal), the variance can be unknown (estimated from data), and the coverage guarantee is exact (non-asymptotic).

Growth rate comparison

Method	Half-width rate
Fixed-sample CI	$\frac{\sigma}{\sqrt{n}} \cdot z_{\alpha/2}$
Normal mixture CS	$\frac{1}{n}\sqrt{(n\sigma^2+\nu)\log\frac{n\sigma^2+\nu}{\nu\alpha^2}}$
LIL-rate CS	$\frac{\sigma}{\sqrt{n}} \cdot \sqrt{2\log\log(n\sigma^2)}$

The Normal mixture CS is wider than the fixed CI by a factor of $\approx \sqrt{\log n}$ (the price of peeking). It is slightly wider than the LIL-rate CS asymptotically, but **tighter in practice** over the 2–3 orders of magnitude of sample sizes relevant to A/B testing.

Why Eppo chose this framework

1. **Anytime-valid:** Experimenters will peek — the guarantee must hold at every moment. ✓
2. **No known variance required:** Real experiments have unknown, potentially changing variance. ✓
3. **No distributional assumption:** Metrics like revenue or session length are far from Normal. ✓
4. **Tunable tightness:** Different experiments have different expected durations. ✓
5. **Produces a confidence interval:** Experimenters need to report “the lift is +3% ± 1.5%”, not just “significant / not significant.” ✓

Key Takeaway

Key concepts: confidence sequences (uniform coverage over all times), the Normal mixture boundary $u(v) = \sqrt{(v+\nu)\log\frac{v+\nu}{\nu\alpha^2}}$, the tuning parameter ν, the sub-ψ framework, nonparametric validity, the $\sqrt{\log n}$ “price of peeking.”

This is the framework Eppo uses. The boundary formula from this act is the engine inside Eppo's confidence interval.

Act 9 — The Problem Eppo Solves

Intuitive Explanation

Intuitive Explanation

A real A/B testing platform faces three problems simultaneously:

1. **The peeking problem** (Act 0): Product managers check results daily. Classical tests break under repeated checking.
2. **High variance:** Users are wildly different. Some buy €0, others buy €500. This noise drowns out the signal — a 2% lift buried under 200% noise. Experiments take weeks to reach significance.
3. **Heterogeneous users:** Not all users respond the same way. Heavy spenders, occasional browsers, mobile vs. desktop — all mixed together. Covariates (pre-experiment behaviour, device type, country) carry information that can separate signal from noise.

Signal vs Noise

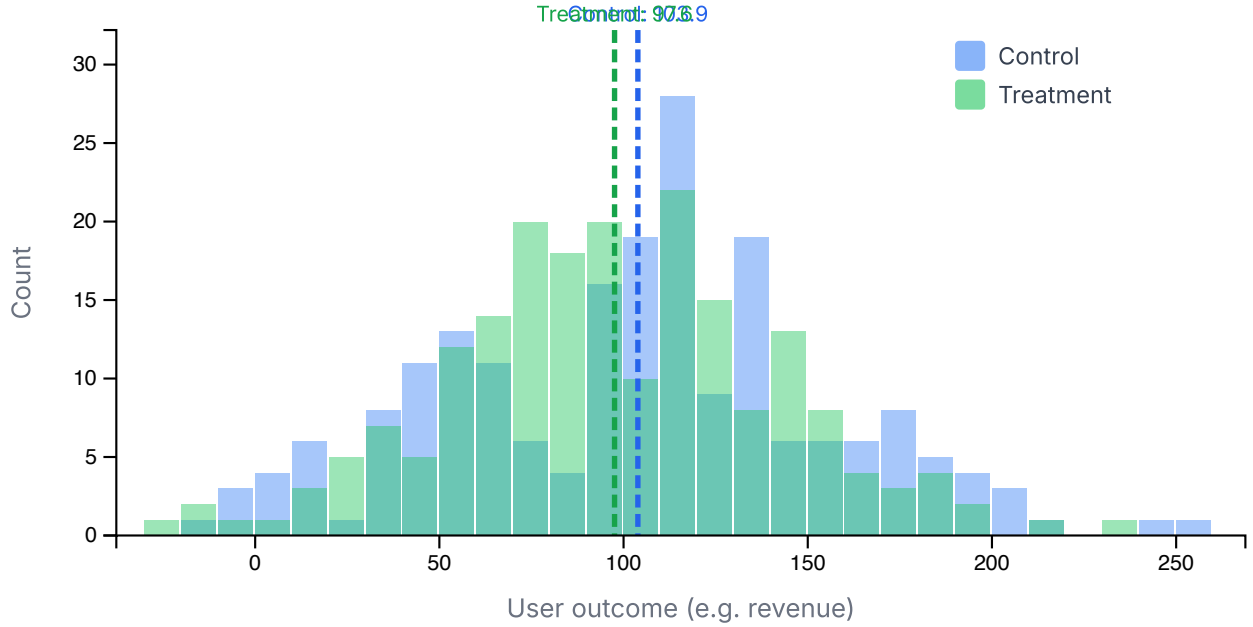
Blue = control, green = treatment. Try increasing noise — the distributions overlap and the signal disappears.

Re-draw

True effect: 2

Noise (σ): 50

Users per group: 200



0.040	-6.33	±9.51	Not significant
Signal / Noise	Observed diff	95% CI half-width	p < 0.05?

The core problem: In real A/B tests, the true effect is tiny (e.g. +2%) while user-level noise is enormous (σ ≈ 200%). This is why experiments need thousands of users and why variance reduction (Act 11) is so valuable — it narrows the distributions, making the signal visible sooner.

What Eppo wants:

- **Anytime-valid inference:** Peek as often as you like; the error guarantee holds.
- **Faster experiments:** Use pre-experiment data to reduce noise, shrinking the confidence interval.
- **Simple outputs:** Report a confidence interval for the *relative lift* ("+3% ± 1.5%") that experimenters can act on immediately.

Acts 0–8 built each ingredient. The next three acts show how Eppo assembles them into one pipeline.

Key Takeaway

Three problems, three solutions:

Problem	Solution	Where we learned it
Peeking inflates errors	Sequential CI (Howard et al.)	Acts 5, 8

Problem	Solution	Where we learned it
High variance / slow experiments	Regression adjustment (CUPED)	Act 11
Need interpretable output	Relative lift with CI	Act 12

Act 10 — The Eppo Pipeline, Step by Step

(Schmit & Miller, 2024)

Intuitive Explanation

Think of Eppo's pipeline as a factory with six stations. Each station does one job. Data enters raw and exits as a trustworthy confidence interval.

`Randomise` → `Collect` → `Adjust` → `Estimate` → `CI` → `Decide`

Step 1: Randomise

Each user arriving at the website is randomly assigned to **control** (existing experience) or **treatment** (new feature). Randomisation ensures the groups are comparable — any difference in outcomes is caused by the treatment, not by pre-existing differences.

Step 2: Collect data over time

As users interact with the product, two quantities are recorded for each user:

- Y_i: the **outcome** during the experiment (e.g. purchase amount, clicks, session length).

- X_i: a **pre-experiment covariate** — the same metric from before the experiment started.

Step 3: Regression adjustment (noise removal)

This is where Eppo *generalises* the CUPED idea from Deng et al. (2013).

For each group **separately**:

1. Fit a regression model predicting Y from the pre-experiment covariates X.
2. Compute the **adjusted outcome** (residual) for each user:

$$Y_i^* = Y_i - \hat{f}_g(X_i)$$

"Predict what each user would have done based on their history, then subtract that prediction. What's left is the part you *couldn't* have predicted — and that's where the treatment effect hides."

Step 4: Estimate the treatment effect

Compute the adjusted group means:

$$\hat{\mu}_0(t) = \frac{1}{n_0(t)} \sum_{i \in \text{control}} Y_i^*, \qquad \hat{\mu}_1(t) = \frac{1}{n_1(t)} \sum_{i \in \text{treatment}} Y_i^*$$

The estimated treatment effect and the **relative lift**:

$$\text{Relative lift}(t) = \frac{\hat{\tau}(t)}{\hat{\mu}_0(t)} = \frac{\hat{\mu}_1(t)}{\hat{\mu}_0(t)} - 1$$

If control averages €100 and treatment averages €103, the relative lift is $103/100 - 1 = 0.03 = +3\%$.

Step 5: Estimate the variance

Compute the standard error of $\hat{\tau}(t)$ from the **residuals** Y_i^*:

$$\hat{\sigma}^2_{\hat{\tau}}(t) = \frac{s_0^2(t)}{n_0(t)} + \frac{s_1^2(t)}{n_1(t)}$$

The variance is estimated from the data — not assumed known. This is a key difference from the mSPRT (Act 7), which required known variance. Howard et al.'s framework (Act 8) permits this.

Step 6: Construct the sequential confidence interval

Plug the estimated treatment effect and its standard error into the Normal mixture boundary from Act 8:

$$\mathrm{CI}(t) = \hat{\tau}(t) \pm \hat{\sigma}_{\hat{\tau}}(t) \cdot \sqrt{\frac{n+\nu}{n} \cdot \log \frac{n+\nu}{\nu\alpha^2}}$$

where $n = n_0(t) + n_1(t)$ is the total sample size and $\nu = M \cdot \hat{\sigma}^2$ is the tuning parameter.

Step 7: Decide

What you see	Colour	Action
CI entirely above 0	Green	Positive effect. Ship the feature.
CI entirely below 0	Dark grey	Negative effect. Revert.
CI crosses 0, experiment ongoing	Grey	Inconclusive. Keep collecting.
CI crosses 0, experiment ended	Grey	No significant effect. Decide pragmatically.

Because the CI is a confidence sequence (Act 8), **this decision is valid at whatever time it is made.** No need to pre-specify an analysis time.

Try It: Interactive Pipeline

EPPO Pipeline Simulation

Watch the full Eppo pipeline in action: randomise users, collect data with CUPED adjustment, build sequential confidence intervals, and see when a decision is reached.

True effect size: 0.15

Max sample size per group: 10,000

-0.3 (harm) 0 (null) 0.5 (strong) 1,000 500,000 1,000,000

Run Pipeline

Key Takeaway

The Eppo pipeline in one sentence: Randomise → collect Y and covariates X → regression-adjust (remove predictable noise) → estimate lift and variance → wrap in a sequential CI → decide when CI excludes zero.

Act 11 — Variance Reduction via Regression Adjustment

Intuitive Explanation

Intuitive Explanation

Imagine weighing yourself before and after a meal. If every person in the study has a different baseline weight (50 kg to 120 kg), the raw "post-meal weight" is extremely variable. But the *change* (before → after) is much more consistent — a meal adds roughly the same weight regardless of the person.

Regression adjustment does the same thing. Each user's outcome Y is partly explained by their pre-experiment behaviour X (their "baseline weight"). The regression model predicts that baseline. Subtracting the prediction leaves the *change* — the part due to the treatment, not the person.

The result: less noise, same signal, tighter confidence intervals, faster decisions.

Variance Reduction: Raw vs CUPED

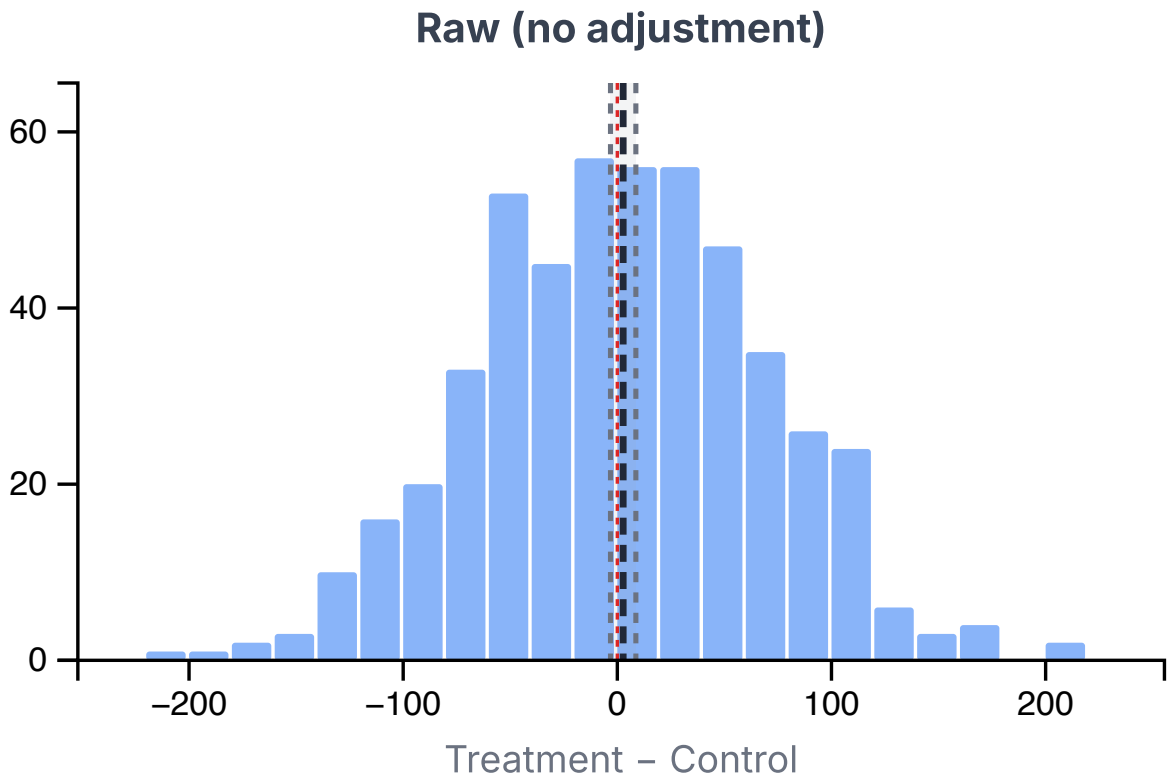
Left: raw differences. Right: CUPED-adjusted. Same data, tighter distribution.
Dashed lines = 95% CI for the mean.

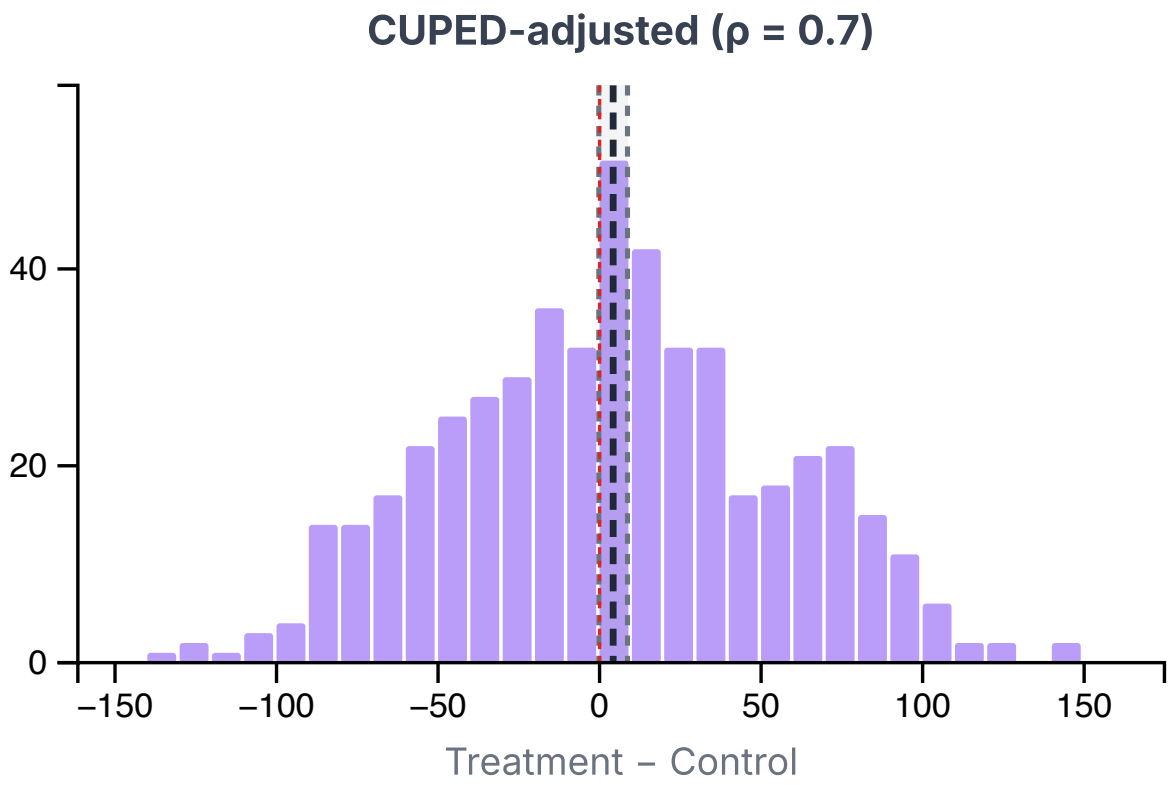
Re-draw

Correlation ρ: 0.70

True effect: 3

Users per group: 500





±5.92	±4.44	43.8%	1.8×
Raw CI width	Adjusted CI width	Variance reduced (theory: 49.0%)	Effective sample multiplier

Mathematical Formulation

From CUPED to generalised regression adjustment

Recall CUPED (Deng et al., 2013):

$$Y^* = Y - \theta(X - \mathbb{E}[X])$$

with optimal $\theta = \mathrm{Cov}(Y, X)/\mathrm{Var}(X)$ and variance reduction $\mathrm{Var}(Y^*) = \mathrm{Var}(Y)(1 - \rho^2)$, where $\rho = \mathrm{Cor}(Y, X)$ is the correlation coefficient.

Eppo generalises this in two ways:

1. **Nonlinear models:** Instead of a single linear coefficient θ, fit a full regression model that can capture nonlinear relationships.
2. **Separate models per group:** Fit $\hat{f}_{\text{control}}(X)$ and $\hat{f}_{\text{treatment}}(X)$ independently.

$$Y_i^* = Y_i - \hat{f}_g(X_i)$$

> "Take what user i actually did. Subtract what the model predicted they would do based on their history. What's left is the residual — the *unpredictable* part."

Why this reduces variance

$$\mathrm{Var}(Y^*) = \mathrm{Var}(Y - \hat{f}(X)) = \mathrm{Var}(Y)(1 - R^2)$$

where R^2 is the coefficient of determination — the fraction of variance in Y explained by X.

R^2	$1 - R^2$	Variance remaining	Traffic multiplier
0.10	0.90	90%	1.1×
0.25	0.75	75%	1.3×

R^2	$1 - R^2$	Variance remaining	Traffic multiplier
0.50	0.50	50%	2×
0.60	0.40	40%	2.5×
0.75	0.25	25%	4×

With $R^2 = 0.50$, the regression explains half the variance. The remaining noise is halved — equivalent to running the experiment with twice as many users.

Critical Safety Rule

The covariate X must satisfy $\mathbb{E}[X^{(\text{treatment})}] = \mathbb{E}[X^{(\text{control})}]$. This is automatically guaranteed when X comes from the *pre-experiment* period (randomisation has not yet occurred).

NEVER use post-treatment covariates. Deng et al. demonstrated that using an in-experiment covariate produced **directionally opposite conclusions**, because the treatment itself changed X.

Why adjustment does not break the sequential guarantee

1. Covariates come from the pre-experiment period (independent of treatment).
2. Regression adjustment changes only the variance, not the mean treatment effect.
3. The sub-Gaussian condition (Act 8) is satisfied by the adjusted residuals.
4. Ville's inequality applies to the resulting supermartingale.

The adjustment is like using a better measuring instrument — more precise, but measuring the same thing. The sequential guarantee depends on the structure of the martingale, which is preserved.

Key Takeaway

Key concepts: regression adjustment as generalised CUPED, separate models per group, variance reduction factor $1 - R^2$, the safety rule (pre-experiment covariates only), adjustment preserves the sequential guarantee.

Act 12 — Sequential Confidence Intervals in Eppo

Intuitive Explanation

Intuitive Explanation

A standard 95% CI says: “Right now, the lift is +3% ± 1.5%.” But that guarantee is only valid at this exact moment. If you checked yesterday, and you check again tomorrow, the guarantee has already been violated.

Eppo's sequential CI says: “The lift is +3% ± 2.1%, and this has been valid *at every moment since the experiment started*.”

The price: the interval is wider — the “± 2.1%” instead of “± 1.5%”. This is the **cost of peeking**. But it is a small price for the freedom to check whenever you want and stop as soon as you are convinced.

The Formula, Revisited

From the pipeline (Act 10, Step 6):

$$\mathrm{CI}(t) = \hat{\tau}(t) \pm \hat{\sigma}_{\hat{\tau}}(t) \cdot \underbrace{\sqrt{\frac{n+\nu}{n} \cdot \log \frac{n+\nu}{\nu\alpha^2}}}_{\text{sequential multiplier}}$$

Compare with a standard CI:

$$\mathrm{CI}_{\text{classical}}(t) = \hat{\tau}(t) \pm \hat{\sigma}_{\hat{\tau}}(t) \cdot \underbrace{z_{\alpha/2}}_{\approx 1.96}$$

The only difference is the **sequential multiplier** replacing the constant 1.96.

How the sequential multiplier behaves

n	$m(n)$ (example)	$z_{\alpha/2}$	Interpretation
100	≈ 3.8	1.96	Early: sequential CI is ~2× wider
1,000	≈ 2.8	1.96	Mid-experiment: gap narrowing
10,000	≈ 2.3	1.96	Late: only ~17% wider
100,000	≈ 2.1	1.96	Approaching classical, never quite reaching it

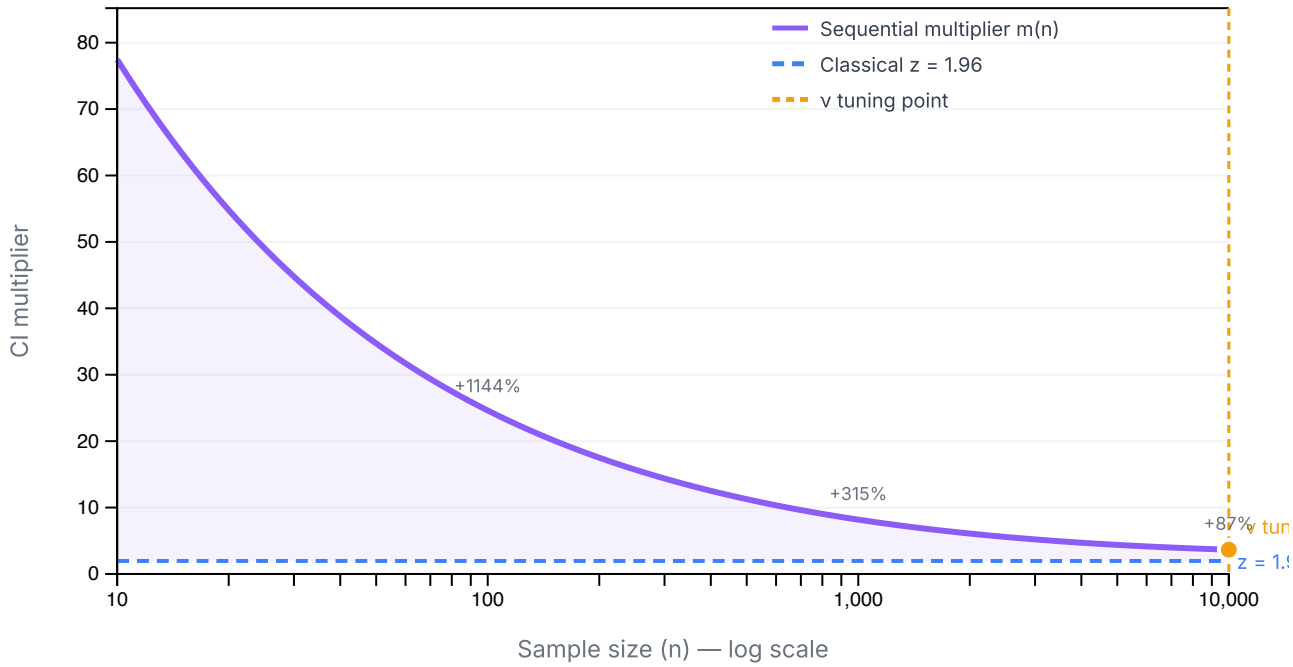
The sequential multiplier starts high and shrinks toward $z_{\alpha/2}$ as more data arrives. The gap — the "price of peeking" — decreases but never fully closes. For most practical experiments (thousands to hundreds of thousands of observations), the price is modest: 10–40% wider than a classical CI.

Sequential Multiplier m(n)

The purple curve shows how the sequential CI multiplier approaches — but never reaches — the classical 1.96. The shaded area is the "price of peeking."

v tuning: 1.0× planned N → tightest at n = 10,000

Max sample size: 10,000



24.62	8.18	3.66

m(100) — 1156% wider than classical

m(1,000) — 317% wider than classical

m(10,000) — 87% wider than classical

The role of ν and M

Schmit & Miller set $\nu = M \cdot \hat{\sigma}^2$, where M is the **expected total sample size**.

If...	Then...
M is too small (underestimate traffic)	CI is tight early but wide late
M is too large (overestimate traffic)	CI is wide early but tight late
M matches actual traffic	CI is tightest around the planned end date

Setting M is like aiming a spotlight: you choose where the beam is brightest. Aim at the expected experiment end, and you get the tightest CI right when you need to make the decision.

What Eppo Actually Uses (Summary)

- **Framework:** Howard et al. (2021) confidence sequences (not the mSPRT of Johari et al. 2017).
- **Boundary:** The Normal mixture boundary $u(v) = \sqrt{(v+\nu)\log\frac{v+\nu}{\nu\alpha^2}}$.
- **Tuning:** $\nu = M\hat{\sigma}^2$, where M is the expected total sample size.
- **Variance:** Estimated from data (not assumed known).
- **Noise reduction:** Generalised CUPED via per-group regression adjustments.
- **Estimand:** Relative lift $\mu_1/\mu_0 - 1$.
- **Decision:** Stop when CI no longer crosses zero, valid at any time.

Act 13 — What If You Don't Have Eppo?

Intuitive Explanation

You want to check your experiment every Monday. If you use a standard 95% CI each time, you'll get false positives far too often (Act 0).

The simplest fix: **be stricter at each check.** If you plan to peek 4 times, demand stronger evidence each time. The three methods below differ in *how* they distribute that strictness across the peeks.

Think of it as a budget. You have a total error budget of $\alpha = 5\%$.

- **Bonferroni** distributes the budget equally: α/K per peek.
- **Pocock** distributes it more cleverly (using the correlation between test statistics at different peeks), but still equally across peeks.
- **O'Brien–Fleming** front-loads the budget: almost no spending early, most of it saved for the final analysis.

Eppo's approach (Acts 8–12) doesn't need a fixed budget at all — it works for *any* number of peeks, continuously.

Method 1: Bonferroni Correction (Simplest)

Before the experiment starts, commit to checking exactly K times. At each check, use a significance level of α/K instead of α.

$$\mathrm{CI}_k = \hat{\tau}(t_k) \pm \hat{\sigma}_{\hat{\tau}}(t_k) \cdot z_{\alpha/(2K)}$$

K (peeks)	α/K	$z_{\alpha/(2K)}$	CI multiplier vs. 1.96
1	0.050	1.96	1.00×
2	0.025	2.24	1.14×
4	0.0125	2.50	1.28×
12	0.00417	2.87	1.46×
52	0.00096	3.29	1.68× (weekly for a year)

Method 2: Pocock Boundaries (1977)

Like Bonferroni, pre-specify K equally-spaced analysis times. But instead of Bonferroni's crude α/K split, use a constant critical value c_P that accounts for the correlation between test statistics at different peeks.

K	c_P (Pocock)	$z_{\alpha/(2K)}$ (Bonf.)	Improvement
2	2.18	2.24	3% tighter
4	2.36	2.50	6% tighter
12	2.60	2.87	9% tighter
20	2.69	3.02	11% tighter

Method 3: O'Brien–Fleming Boundaries (1979)

Instead of a constant threshold, use a threshold that *decreases* over time. At the first peek, demand very strong evidence; at the last peek, demand nearly the same evidence as a classical test.

$$c_k^{\text{OBF}} \approx z_{\alpha/2} \cdot \sqrt{K/k}$$

Peek k (of $K = 4$)	c_k^{OBF}	c_P (Pocock)	$z_{\alpha/2}$ (classical)
1 (25% of data)	4.05	2.36	1.96
2 (50% of data)	2.86	2.36	1.96
3 (75% of data)	2.34	2.36	1.96
4 (100% of data)	2.02	2.36	1.96

O'Brien–Fleming is extremely conservative early (threshold 4.05 at the first peek) but barely penalises the final analysis (2.02 vs. 1.96 — only 3% wider). This is the most efficient approach: if you run the experiment to completion, you pay almost nothing for the option to peek.

Head-to-Head Comparison

	Bonferroni	Pocock	OBF	Eppo
Pre-specify K?	Yes	Yes	Yes	No
Threshold type	Constant	Constant	Decreasing	Continuous
Valid between peeks?	No	No	No	Yes
Variance reduction?	Manual	Manual	Manual	Built-in
Implementation	1 line	Lookup table	Formula	Platform

Which method should you use?

- **No platform, minimal effort:** Bonferroni. Commit to K peeks. Replace 1.96 with $z_{\alpha/(2K)}$. Done.
- **No platform, moderate effort:** O'Brien–Fleming. Nearly as efficient as Eppo at the final analysis, but requires pre-specifying K.
- **Need continuous monitoring or variance reduction:** Use a platform like Eppo.

The bottom line: even the crudest correction (Bonferroni) is *vastly* better than naive peeking. If you are checking results without any correction, you should start with Bonferroni *today*.

Try It: Compare Methods

Head-to-Head Method Comparison

Compare how Bonferroni, Pocock, O'Brien–Fleming, and Confidence Sequences behave on the same data. All methods use 12 equally-spaced peeks.

True effect: 0.20

Sample size per group: 10,000

Run Comparison

Historical Note

Stuart Pocock (1977) and **Peter O'Brien & Thomas Fleming** (1979) developed these group sequential methods for clinical trials, where stopping a trial early can save lives. Their work sits between Wald's SPRT (1945) and Howard et al.'s continuous framework (2021) in the historical timeline. The key insight of all three: if you want to peek, you must pay — but the price is negotiable.

Summary: The Through-Line

Act	Object	Represents	Key idea
0	Repeated p-values	The peeking problem	Checking repeatedly inflates FP
1	Random walk S_n	Position	SD grows as $\sqrt{n}$
2	Winnings M_n	Gambler's fortune	Martingale; Doob: $\mathbb{E}[M_\tau] = M_0$
3	Likelihood ratio Λ_n	Evidence against H_0	Comparing hypotheses flip by flip
4	Λ_n under H_0	LR is a martingale	The cancellation proof
5	Markov + Ville	Anytime-valid guarantee	$\mathbb{P}(\Lambda_n \text{ ever} \geq 1/\alpha) \leq \alpha$
6	SPRT	Wald's decision rule	Optimal but requires known δ
7	Λ_n^H (mixture LR)	Robust evidence	Robbins' mixtures → Johari's mSPRT
8	Confidence sequence	Modern framework	Howard: nonparametric, estimated σ^2
9	Peeking vs. variance	Three problems	Peeking, high variance, heterogeneous users
10	Eppo pipeline	Step-by-step factory	Randomise → adjust → CI → decide

Act	Object	Represents	Key idea
11	Regression adjustment	Noise removal	$\mathrm{Var}(Y^*) = \mathrm{Var}(Y)(1 - R^2)$
12	Eppo's sequential CI	Final output	Sequential multiplier; decide anytime
13	DIY alternatives	Simple approximations	Bonferroni, Pocock, OBF vs. Eppo

The core message: All fourteen acts build on the same mathematical structure — non-negative (super)martingales and Ville's inequality. The anytime-valid guarantee flows from a single elegant fact:

> *A fair game cannot be beaten by choosing when to quit.*

The evolution across eight decades:

Ville (1939) → Wald (1945) → Robbins (1970) → Johari (2017) → Howard (2021) → Eppo (2024)

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